

ALLEN M. SWACKHAMER

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EDUCATION/QUALIFICATIONS:

- B.B.A. Infrastructure Assurance, University of Texas at San Antonio, 2010
- B.B.A. Information Systems, University of Texas at San Antonio, 2010

SUMMARY OF RELEVANT EXPERIENCE:

COMPUTER SOFTWARE

- 10+ years of cyber security development operations experience
- Proven ability to develop and mature cyber security programs, with a strong focus on threat defense through red team collaboration and detection engineering. Experienced in evolving incident response capabilities.
- Extensive knowledge of Python with experience in go, bash, C/C++, ASM, and Java
- Static and dynamic analysis tools / frameworks for malware analysis including IDA Pro, Windbg, Immunity, Volatility and Cuckoo

COMPUTER AND NETWORK SECURITY

- Enumerating network based vulnerabilities and penetration testing using open source tools and frameworks
- Working with Windows memory captures utilizing open source memory analysis frameworks to identify malware artifacts
- Extensive knowledge in network forensics tools utilizing and developing network protocol analyzers

DEVELOPMENT OPERATIONS

- Designed and implemented comprehensive automated solutions for testing, building and deploying open-source tools as well as custom-developed applications across diverse production and staging environments, including cloud and bare metal
- Developed custom code and tooling for Whitebox hardware deployments using modern Infrastructure as Code (IaC) solutions and built platforms from the ground up, starting with the bare metal acquisition process
- Created robust monitoring and alerting solutions for bare metal to custom code using modern open source logging, monitoring and alerting frameworks from the Cloud Native Computing Foundation (CNCF) and others
- Led testing teams for large-scale projects designed to run on tens of thousands to a million computers
- Configured, installed and administered cyber ranges that replicated large enterprise networks

PATENTS

- Network security analysis and malware detection using multiple types of malware information (11134090 & 10735442)
- Enriching malware information for use with network security analysis and malware detection (10972484)

EXPERIENCE:

PRINCIPAL PRODUCT INCUBATION ENGINEER, SIMSPACE

2022-2023

AUTOMATION / RELEASE ENGINEERING

- Led the deployment, administration and monitoring of the Next Generation Simspace Platform
- Built multiple architecture Docker containers, deployed to Google's Kubernetes Platform (GKE) with Gitlab CI/CD, migrated to GitHub and utilized Concourse CI and Argo CD for GKE deployment
- Monitored and logged GKE with Loki and Prometheus Community Operator using Alertmanager to send alerts to slack when custom and select community rules were triggered

SOFTWARE DEVELOPMENT

- Created multiple proof of concept tools and products building on to the product line and existing products
- Developed a way to pull asset inventory from PCAP using Bro scripts and parsing Bro logs from packet captures and active scans, with the ultimate goal of emulating what was detected in a cyber range
- Utilized K8s to emulate multiple hosts using the MetalLB Kubernetes load balancer

PRINCIPAL ENGINEER, CYBER FUSION CENTER, TARGET

2016-2022

MALWARE ANALYST / REVERSE ENGINEER

- Developed Cyber Threat Intelligence capabilities to track threat actors across the malware threat landscape

- Expanded and created a big data product that processed files from various file feeds from VirusTotal to Abuse.ch to collect open source intelligence on the static files and pull a subset of those files to perform more in depth analysis to generate IOC's for Detection Engineers
- Obtained multiple patents for Target based on the capabilities and usage of the Intel Tools intelligence collection capability
- Worked with the Cyber Threat Intelligence team to perform in depth malware reverse engineering using static disassembly and dynamic analysis tooling

PLATFORM ENGINEERING / MONITORING & ALERTING

- Built, from the ground up, a multiple rack cluster of computers originally starting small with ESXi / oVirt and moving to OpenStack to support more clients that wanted to interact with the datasets, or perform direct static/dynamic analysis. Eventually, this platform moved to Vanilla Kubernetes. Both OpenStack and Kubernetes were deployed utilizing Ansible with a combination of community and custom configuration.
- Hosted a wide range of products and operators on this remote platform, including: Kubernetes, OpenStack, Ceph for block and object storage, Elasticsearch/Logstash/Kibana, and Kafka
- Maintained and hosted a branch of the Strelka file analysis framework that was specifically developed and deployed for CTI and RE's on the intrusion response team
- Automated the deployment of most infrastructure using Drone CI/CD which would deploy K8s manifests and perform build and push operations for Docker images representing the application stack
- Created a monitoring and alerting system using Prometheus Operator utilizing Alertmanager to alert on any failures in the entire application stack from hard disk SMART failures to Kubernetes container runtime crash looping

SIEM DEVELOPMENT

- Led the backend development of the replacement SIEM that Target used which consisted of a Logstash/Kafka/Elasticsearch pipeline that consumed logs from Strelka and a massive flow (100k EPS+) of Bro logs from all of the network sensors
- Utilized Chef to deploy Elasticsearch onto large Dell Blades that consisted of thousands of CPU cores and multiple TB's of RAM. Eventually moved to utilizing ansible to deploy Elasticsearch using Docker and Systemd unit scripts
- Developed logstash filters to parse bro and other log types for ingestion into Elasticsearch as well as created ES mappings for these datasets
- Utilized internal OpenStack and HEAT templates to automate the deployment of the stage network

ASSOCIATE, STRATEGIC INNOVATION GROUP, BOOZ ALLEN HAMILTON

2010–2016

MALWARE ANALYSIS DEVELOPMENT AND INCIDENT RESPONSE

- Conducted in-depth incident response for a commercial client which represented a major breach of highly sensitive data. Utilized open source tools and libraries to automate the creation of accurate timelines of events into a super timeline. Gathered data from a multitude of sources to make informed conclusions on if the adversary was still present in the network. Built custom scripts to detect additional IP's related to the attack based on SSL signatures
- Consulted for multiple government and commercial companies on how to strengthen their network and performed in depth risk analysis
- Performed penetration testing for multiple commercial and government clients ranging from physical penetration testing to in depth large scope targeting all operating environments internal and external as well as wireless.
- Developed a malware analysis to feed VirusTotal originated binaries into dynamic sandboxes to populate an Elasticsearch database
- Lead development of a malware analysis platform utilizing both static and dynamic analysis for a DoD client
- Integrated in open source memory forensic toolkits that allowed end analysts to quickly analyze Windows executables
- Found hundreds of malware artifacts on production enterprise networks through statistical analysis of data provided by in house host based collection tools

QUALITY ASSURANCE / RELEASE ENGINEERING

- Led a small team of three testers in developmental testing of cyber defense tools for the United States Air Force.
- Cut releases that were delivered to customers through CI/CD pipeline utilizing Atlassians Bamboo for a wide range of products on the defensive side and specialized embedded projects on the offensive side.
- Designed, built and administered a developmental test network utilizing virtual machine technologies and continuous integration software to automate testing of tool development
- Lead developmental testing for major cyber defense tool products that successfully rolled out to tens of thousands of clients
- Reported and fixed numerous major software security vulnerabilities during the development of cyber defense tools

SOFTWARE DEVELOPMENT

- Developed a flexible modular application to detect unknown malicious applications running on hosts using live memory analysis techniques for Windows systems.
- Innovated new techniques to detect packed binaries using custom heuristics by analyzing processes running on a host
- Extended host-based collection tools to report on possible memory regions where malware could be present